THE MAGAZINE OF EFFECTIVE COMPARTMENTATION

Life Safety DIGEST

WINTER 2024

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Keeping the Lead on Safety

Barriers are the Heart of Compartmentation

All Ceiling Radiation Dampers are not EQUAL

Special Locking Arrangements vs Normal Locking Arrangements -What's the Difference?

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LIFE SAFETY DIGEST

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EDITOR'S MESSAGE

FCIA CELEBRATES 25 YEARS – "A LOT DONE, A LOT TO DO"

Last November in Palm Springs, California the FCIA celebrated its' 25th Anniversary. Twenty-five years of focusing on the 'DIIM'™ of Firestopping to ensure that installed fire-resistance-rated and smoke-resistive assemblies perform as designed, which means safer buildings protecting occupants and property.

The FCIA's 'DIIM'™ Story for Firestopping did not happen by magic. It was purposefully established by FCIA's 8 member Steering Committee in 1998, refined at FCIA's 1st Conference in 1999 – and further developed as the 25 years have occurred.

To get the 'DIIM'™ built, FCIA's Board developed the strategy. It was fulfilled by FCIA Committees, whose members, supported by staff, 'got stuff done'. FCIA attended and spoke at CSI/CSC Specification Association Conferences, International Code Council and NFPA Expo's & Building and Fire Code Development Hearings, participated at the ASTM and UL Standards Development Processes…presenting and advocating for the FCIA's 'DIIM'™ Firestopping Story. What started in the USA, expanded first to Canada, then the United Arab Emirates, Qatar, and now worldwide. As a result, FCIA Membership has grown from 8 in 1998 to 375 plus companies in 2024.

FCIA's 1st President – Aedan Gleeson - mentioned, "we want this effort to outlive all of our time here". As FCIA celebrates its 25th anniversary, we are thankful that it has, and will outlive our time with this group. We're thankful for the FCIA Members, who fund the work we do through membership in the association, volunteer board and committee actions and staff that has made FCIA what it is today.

On behalf of the 8 leaders who were on the steering committee, thanks for making the FCIA's 'DIIM'™ story your story – and taking it forward. We know that FCIA will be guided over the next 25 years by great leadership and staff, and in the manner the next group determine serves the 'DIIM'™ the best.

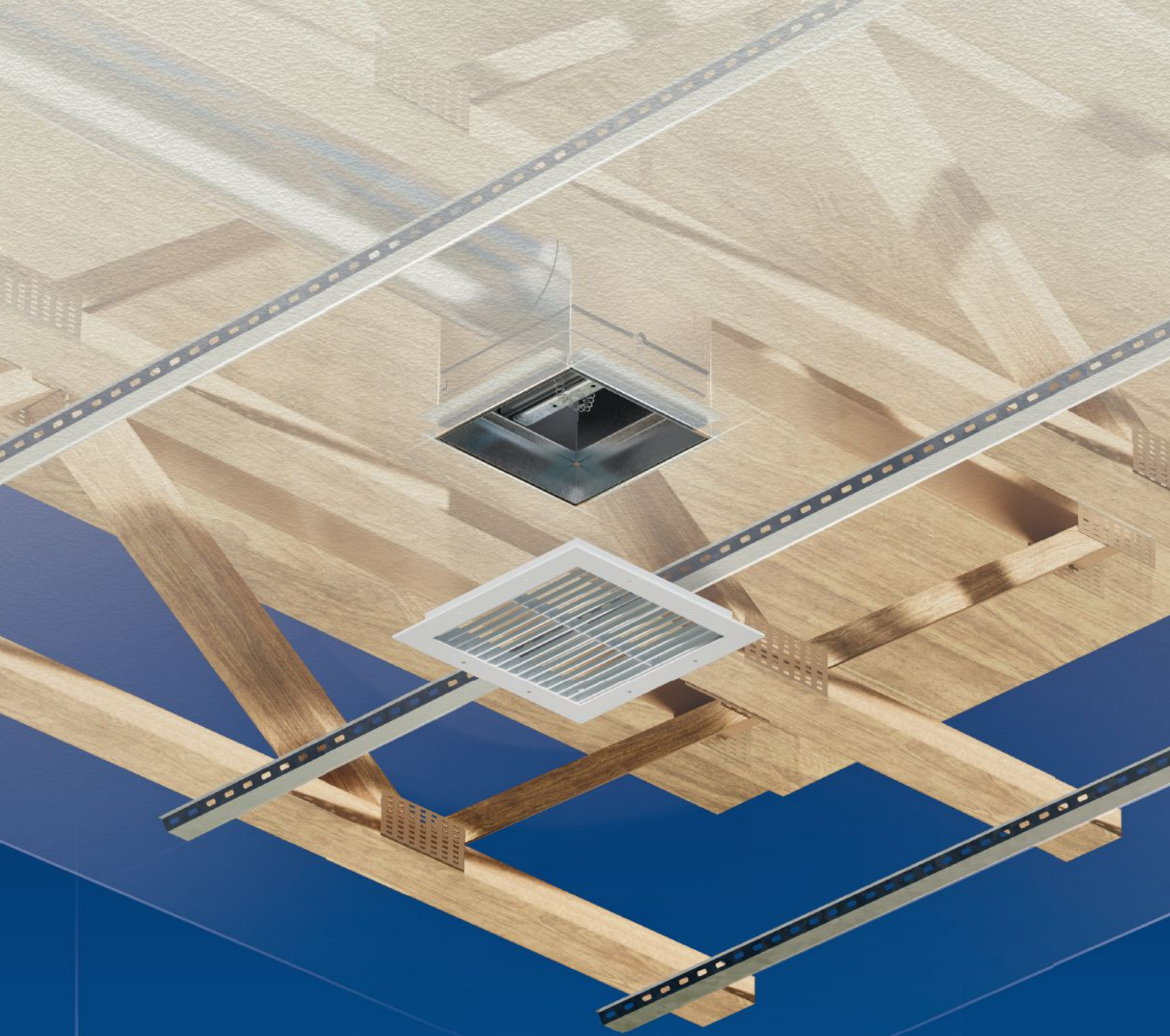
Sincerely,



William McHugh
Publisher



FCIA's 1st FIC Conference, January 1999, developing the 'DIIM'™ Story at UL's Chicago area HQ. FCIA Photo



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FIRE-RESISTANCE & FIRESTOPPING IN UAE

– AN INTERVIEW WITH ENG. PRAMOD CHALLA, EXPERT - FIRE AND LIFE SAFETY ENGINEERING AT SHARJAH CIVIL DEFENCE AUTHORITY, UNITED ARAB EMIRATES.

COMPILED FROM AN INTERVIEW CONDUCTED BETWEEN MR. ABHISHEK CHHABRA AND ENG. PRAMOD CHALLA

FCIA's Director, International Initiatives, Abhishek Chhabra had the opportunity to visit with Eng. Pramod Challa, Sharjah UAE's Expert – Fire and Life Safety Engineering, to learn more about fire-resistance in the United Arab Emirates (UAE). Below is a summary of the conversation.

ABOUT ENG. PRAMOD & UAE CIVIL DEFENCE

I am an instrumentation engineering graduate from Karnataka, India. I began my career with Saudi Arabia Electricity's Preventive Safety Department in the late 1990's and worked for 8 years in the power industry.

In 2008, I moved to Dubai UAE Civil Defence, serving as a Senior Engineer until late 2023. Since then, I have been with Sharjah UAE Civil Defence, in the role of Fire and Life Safety Expert.

Throughout my tenure in UAE Civil Defence, with enthusiastic management support, I have played a pivotal role in establishing project approval procedures, starting from the design submission process, submission format, developing an online portal called 'e-engineering', and writing of the UAE Fire and Life Safety Code of Practice. My work also includes streamlining testing laboratory and certification body registration, material testing and approval processes, and certifying industry professionals through examinations. It gives me immense satisfaction to have shaped how the fire industry interacts with UAE Civil Defence.

HOW HAS FIRESTOPPING GAINED IMPORTANCE IN THE UAE AND HOW HAS THE UAE FIRE AND LIFE SAFETY CODE BROUGHT ABOUT THE CHANGES?

Firestopping has gained significant importance over the years due to its critical role in overall fire safety. Initially, when the UAE Fire and Life Safety Code was first released in 2011, a dedicated section on firestopping was included, though it was acknowledged, industry practices were not fully aligned with stringent standards seen in international codes. While the first version addressed the issue, there was a gap in industry practices, and enforcement wasn't as robust.

By the time the 2018 version of the UAE Fire and Life Safety Code was released, there was an attempt to highlight industry loopholes, such as improper selection of systems, improper onsite installations and lack of standardization.

Under the leadership of Civil Defence, initiatives were launched to educate the industry, focusing on training designers, certifying technicians, and ensuring compliance across various stages of the construction process. Many industry professionals also stepped up, including the Firestop Contractors International Association, International Firestop Council, manufacturers and others contributing through workshops to provide education. All of which have significantly improved firestopping practices in the region.

However, despite the progress, I believe we're still at the early stages of optimizing firestopping quality. There are gaps that need attention, such as designers leaving decisions on firestopping system type selection to field level when it should be at the supervisory or design stage. Quantities of materials being delegated to suppliers is also an issue. Contractors relying on general labour and technicians for installation without strong designated responsible individuals providing supervision is a problem. Finally, lack of consistent, progressive inspections on-site is a problem as well.

Additionally, authorities need to enforce stricter verification of firestopping installation and inspection reports. Addressing these areas will further strengthen fire safety and ensure a more comprehensive approach to firestopping in the region and globally.

HOW CLADDING & FAÇADES FIRE SAFETY WAS BOLDLY DRIVEN BY CHANGES IN THE CODE

Cladding and façade fire safety became a priority in the region following several high-profile façade fires, both locally and globally. Terms such as "flammable core," "combustible curtain wall arrangements," and "firestopping" started gaining serious attention within the industry.

Interestingly, the 2011 UAE Fire and Life Safety Code was relatively silent on this matter, despite these concerns already being addressed in international codes. This left many of us, as authorities, questioning why designers and consultants were not proactively tackling these issues, especially when clear standards existed elsewhere.

The 2018 version of the UAE Fire and Life Safety Code took a bold and decisive step in addressing cladding and façade fire risks. Before many other parts of the world could catch up, the UAE had already set a new benchmark in façade fire safety.

The updated UAE Fire and Life Safety Code introduced systematic and multi-level testing of façade system components and assemblies as complete systems. It also mandated progressive façade report approvals and required materials to be registered with Civil Defence. These measures played a crucial role in minimizing the occurrence of devastating façade fires.

The industry responded by phasing out flammable cores in façade panels, upgrading designs to meet new code requirements, and following stringent testing and approval procedures. Unlike some regions that have opted for a complete ban on cladding panels, the UAE Fire and Life Safety Code since 2016 has taken a more balanced approach, allowing for the use of code-compliant materials and assemblies. This approach not only ensures safety but also maintains architectural flexibility and innovation.

GIVE SOME EXAMPLES OF HOW SOME PARTS OF THE CODE PROGRESSED OVER THE YEARS (PART OF ABOVE)

The UAE Fire and Life Safety Code has made significant strides introducing new chapters and refining regulations to address emerging challenges and improve fire safety. One of the most important advancements in the 2018 edition was the introduction of a dedicated chapter on the Responsibilities of Stakeholders. This new section was pivotal in clarifying and legally binding the roles and obligations of all stakeholders involved in a project, from consultants and contractors to manufacturers, test laboratories, and certification bodies.

In this chapter, the responsibilities of each stakeholder are explicitly outlined, ensuring that everyone involved in the design, construction, and approval process is accountable for maintaining fire safety standards. This change not only provided clarity but also created a legal framework that holds stakeholders responsible for violations, ensuring that negligence or non-compliance could lead to serious consequences, including fines and penalties.



Dubai's skyline, filled with high-rise buildings. FCIA Photos



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SOME KEY AREAS OF PROGRESS INCLUDE:

Multilevel Testing and Certification - As mentioned earlier, façade and cladding systems are now subject to both component-level testing and full system assembly testing. This ensures that all materials used, from the core to the external finishes, meet the highest fire safety standards. The large-scale testing is comprehensive, covering fire-resistance, spread of flame, and curtainwall integrity, among other critical factors.

Clear Accountability - The code assigns specific duties to consultants to ensure designs are fire-safe, to contractors for proper installation of fire-resistance-rated systems, to manufacturers for supplying compliant materials, and to test laboratories and certification bodies for conducting accurate and unbiased testing. By clearly defining these roles, the code eliminated much of the ambiguity that previously existed.

Punishable Actions for Violations

One of the strongest provisions in the 2018 update is the introduction of stringent, legally enforceable penalties for non-compliance. Stakeholders found violating the code or failing to fulfill their responsibilities can face fines, project shutdowns, or even legal action. This accountability has significantly reduced the number of violations and promoted a culture of compliance and safety within the industry.

Legal angle, what empowers the authorities to implement the code. Decree 213

The UAE *Decree 213* further empowers Civil Defence to:

- Approve building designs and materials (such as façades, cladding, and firestopping components) in line with the UAE Fire and Life Safety Code.
- Conduct site inspections to ensure compliance with the code throughout the construction process.
- Certify professionals involved in fire safety roles, ensuring that they are adequately trained and competent.
- Oversee the testing, certification, and approval of fire protection systems and materials before they can be used in the market.

In addition, it provides the authority to introduce new code updates, ensuring regulations evolve with technological advancements and emerging fire safety risks. For instance, the 2018 version of the UAE Fire Code saw a substantial overhaul in response to modern challenges like cladding and façade fire risks, with legal backing from *Decree 213* ensuring these changes were enforceable.

And, upcoming UAE Fire and Life Safety Code of Practice versions will address latest threats such as Lithium-Ion battery fire safety, Artificial Intelligence (AI) in fire safety etc.

SHARE SOME RECOURSES THAT THE INDUSTRY HAS TO REPORT NON-COMPLIANCES WITNESSED IN THE MARKET:

The UAE Civil Defence has always maintained an industry-friendly approach, ensuring that fire safety regulations evolve with input from professionals across the sector. The UAE Fire and Life Safety Code itself was established and developed with active participation from industry experts, with many deviations, revisions and adaptations to international standards being acknowledged and validated by those directly involved. This collaborative spirit continues in the approach to reporting non-compliances within the market as well.

The industry has several key channels to report violations or improper practices regarding fire safety:

Direct Communication with Civil Defence

Responsible stakeholders, including consultants, contractors, manufacturers, and certification bodies, are encouraged to report any non-compliances or violations directly to the Civil Defence authorities within their respective Emirates (Note; an Emirate is like a State, Province). Whether it's a breach of fire safety codes during construction, improper installation of fire systems, or the use of non-certified materials, these concerns can be submitted in writing to the relevant Civil Defence department.

UAE Fire Code Committee - Stakeholders also have the option to reach out directly to the UAE Fire Code Committee. This committee is responsible for reviewing and updating the fire code, and it is receptive to feedback from industry professionals regarding non-compliance or areas where the code may need further clarification. Written submissions to the committee help ensure that violations are addressed, and they often lead to corrective actions or even code revisions where necessary.

Online Portals - Many Emirates have plans of online portals where stakeholders can file complaints or report non-compliant practices. These platforms will be part of the ongoing efforts to streamline communication between the industry and Civil Defence, ensuring a more efficient and accessible way to address fire safety violations.

Civil Defence Inspections - Additionally, Civil Defence conducts routine and random inspections of construction sites and operational buildings. While these inspections are a formal means of ensuring compliance, they also provide an opportunity for stakeholders to raise any concerns they may have observed on-site, further promoting a culture of transparency and safety.

Editors Note: FCIA and Life Safety Digest appreciates all Engineer Pramod Challa has contributed to the UAE's Fire and Life Safety environment. His dedication to pursuing a safer new construction and existing building stock is amazing – and readily seen in the UAE Fire and Life Safety Code of Practice. He continues his work through the Sharjah, UAE Civil Defence office.



WRITTEN BY **JOE RIGGI**

KEEP THE LEAD ON SAFETY

In the dynamic world of industrial operations, ensuring the safety of employees and maintaining production are paramount. Industrial occupancies, encompassing manufacturing plants, chemical facilities, warehouses, and other similar environments, present unique challenges, and risks. Staying ahead of safety requires a proactive and multifaceted approach, integrating regulatory compliance, technological advancements, employee training, and continuous coaching for improvement.

In industrial occupancies, high-risk environments with heavy machinery, hazardous materials, and complex processes are commonly found. These environments can vary widely, from automotive manufacturing plants with automated assembly lines to chemical processing facilities handling volatile substances. Each type of industrial occupancy presents specific hazards and risks, requiring compound safety strategies for both the building owner and manager, as well as the passive fire protection contractor.

Plus, there might be different safety requirements for new construction than in an existing building. Where new construction might not have facility operational issues to deal with, an existing building will have many risks that need to be addressed by the passive fire protection contractor, covering several safety standards.

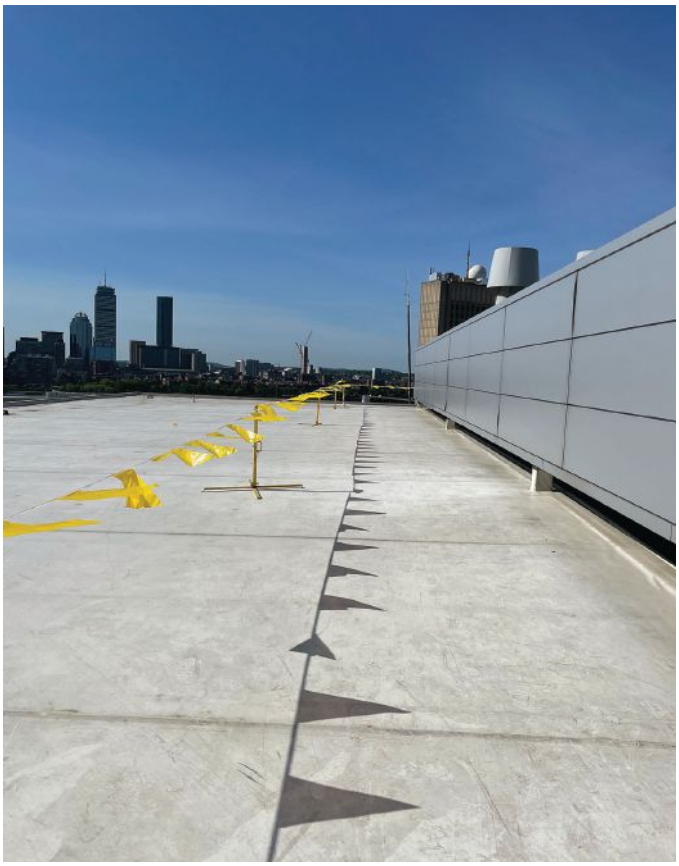
This article focuses on safety in a very broad sense, from a USA perspective. While the safety standards might be similar for each country, we respect that each country adopts its own standards either internally or using standards available throughout the world.

REGULATORY COMPLIANCE AND STANDARDS, REGULATIONS

One of the cornerstones of industrial safety is adherence to regulatory standards and guidelines. In the United States, organizations such as the Occupational Safety and Health Administration (OSHA) set forth regulations to safeguard workers. Compliance with OSHA standards is not merely a legal obligation, but a foundational component of an effective safety program.

OSHA regulations cover a wide range of safety aspects, including machine guarding, hazardous materials handling, fall protection, and emergency preparedness. Staying informed about these regulations and any updates is crucial for ensuring compliance. Regular audits and inspections by the workforce and management can help identify areas where standards may not be met, prompting corrective actions.

Beyond general OSHA regulations, certain industries have additional standards. For example, the National Fire Protection Association (NFPA) provides guidelines for fire safety and egress. Other Standards Development organizations (SDO's), most accredited by the American National Standards Institute (ANSI), offer standards on safety and personal protective equipment (PPE). SDO's such as the American Society of Safety Professionals (ASSP), American Society of Safety Engineers, (ASSE), International Safety Equipment Association (ISEA), and others are involved in developing safety standards. OSHA sets minimum requirements for employers to meet. However, understanding and implementing industry-specific safety standards can further enhance safety – helping bring workers home safe to their families.



Safety inside and outside is critical to workers getting home safe every day. Joe Riggi Photo

INCORPORATING TECHNOLOGICAL ADVANCEMENTS

Technology plays a pivotal role in advancing safety in industrial settings. Innovations in automation, data analytics, and communication tools have the potential to significantly reduce risks and improve safety outcomes.

Automation and robotics can mitigate human exposure to hazardous conditions in factory and warehouse occupancies. For example, robotic arms can handle dangerous chemicals or heavy materials, reducing the risk of injury. They routinely pick items to be packed for shipment. Additionally, automated systems can monitor and control environmental conditions, ensuring that safety parameters are maintained.

While animation and robotics work well on a factory floor, they might not be options for construction projects where working at height and in tight spaces might preclude robot usage. When installing firestopping at the head of wall in a facility with extensive piping and electrical needs, access to the areas for firestop might be quite limited. Robotics might not be able to access those locations where access is minimal.

DATA ANALYTICS AND PREDICTIVE MAINTENANCE:

Leveraging data analytics for predictive maintenance of equipment used to build fire-resistance-rated assemblies in buildings is important – because manufacturers state this in their instructions. Not following maintenance instructions can result in pumps, dispensing devices and power tools being unfit for use at a project site. Maintaining all types of equipment – factory production or construction equipment - can prevent failures before they occur. By analyzing historical data and real-time monitoring, predictive maintenance systems can identify potential issues and schedule maintenance activities proactively, reducing the likelihood of accidents caused by equipment malfunction.

Modern safety management software can provide tools for incident reporting, safety audits, and compliance tracking. When used, the systems streamline processes, making it easier to identify trends, manage risks, and ensure that safety protocols are followed consistently by employees.

EMPLOYEE TRAINING AND ENGAGEMENT:

No safety program can succeed without the active participation and awareness of employees. Training and engagement are crucial for fostering a culture of safety and ensuring that workers are prepared to handle potential hazards. Safety management software might be able to track employee safety training to help ensure employer compliance with OSHA and other safety standards.

Certain building owners mandate types of safety training to meet minimum requirements, and in some cases exceed requirements. OSHA's 10 hour course seems to be a minimum requirement to work on construction jobsites. In addition to the OSHA 10, a 30-hour course is offered, which can be combined with confined space training, personal protective equipment education, toxic and hazardous substance education for workers at construction projects.

The type of education required by OSHA standards might vary if the structure is new or an existing building. Existing buildings might have materials considered by OSHA to present risk(s) to the construction worker. New construction might have egress issues. Education about safe egress in emergency situation(s) is key for construction workers. Knowing the multiple ways out of a building is key to fire and life safety at construction sites where the route out might not be obvious and intuitive.

Conducting regular safety drills and simulations can help employees practice their responses to emergency situations. Exercises can include fire drills, chemical spill response drills, and evacuation procedures. Regular practice ensures that employees are familiar with emergency protocols and can act swiftly in real situations.



Floor holes can be a big risk to safety. Joe Riggi Photo

ENCOURAGING SAFETY CULTURE:

Building a safety culture involves creating an environment where safety is a shared responsibility. Encouraging open communication, recognizing safe practices, and involving employees in safety planning can foster a proactive approach to safety. Employee feedback should be valued and used to improve safety practices.



Occupants participate in safety too. Joe Riggi Photo

Safety for passive fire protection workers in construction or maintaining protection in industrial environments is an ongoing process that requires continuous evaluation and improvement. Regular risk assessments and updates to safety protocols are essential for adapting to new challenges and maintaining high safety standards. When working in industrial occupancies, the passive fire protection workers need to understand safety in both construction as well as the facility specific safety issues that are important for a safe work environment. Below are things to consider when working in various environments.

- **Wall & Horizontal Assemblies**
  - o Ensuring proper installation to prevent the spread of smoke and fire
  - o Inspection to confirm installation in accordance with fire-resistance-rated listing, manufacturers' instructions (or verifying installation in accordance with calculated or analytical methods).
  - o Annual visual inspection for maintaining protection for the building life cycle.

- **Fire Damper Installation, Inspection & Maintenance:**
  - o Ensuring proper installation of fire dampers to prevent the spread of smoke and fire through ducts and ventilation systems.
  - o Regular inspection of fire dampers to confirm compliance with fire safety regulations and to ensure they function properly during an emergency.
  - o Routine maintenance and testing to ensure dampers close securely during a fire, preventing fire from traveling through air vents.
  - o Replacement of any damaged or non-functional dampers to ensure continued protection in fire-risk areas.
  - o Documentation of inspection and maintenance activities for compliance with fire safety standards and regulations.
  - o Wearing fall protection personal protective equipment (PPE) key to safe work at height. Gloves for preventing skin cuts and more.

- **Firestop Systems Installation, Inspection & Maintenance**
  - o Proper installation of firestop sealants, wraps, devices, boards, blocks protecting fire-resistance of breaches in fire-resistance-rated and smoke resistive walls and floors – to provide safe compartments for building occupants and structures.
  - o Proper fall protection PPE working at heights, along with PPE as required by manufacturers safety data sheets.
  - o Inspection to ensure listings and manufacturer's instructions are followed.
  - o Annual visual inspection for maintaining protection for the building life cycle.

- **Fire Door Installation, Inspection & Maintenance:**
  - o Proper installation of fire-rated doors to meet building code and fire code requirements ensure they can withstand high temperatures and prevent fire spread.
  - o Regular inspection of fire doors to check for proper operation, ensuring they close automatically and form a tight seal to limit the spread of smoke and flames.
  - o Maintenance of fire doors to repair any damage such as cracked seals, broken hinges, or faulty locking mechanisms that could compromise their effectiveness.
  - o Ensuring the correct signage and labeling for all fire doors, making them easily identifiable during an emergency.
  - o Training staff to properly maintain clear and unobstructed fire doorways to ensure their functionality during an evacuation.

These practices ensure the integrity of passive fire protection systems, minimizing risk and enhancing safety for building occupants in case of a fire.

Conducting regular risk assessments helps identify potential hazards and evaluate the effectiveness of existing safety measures. Techniques such as Failure Modes and Effects Analysis (FMEA) or Hazard and Operability Study (HAZOP) can provide valuable insights into potential risks and areas for improvement.

Adopting continuous improvement through analyzing incidents and near misses helps prevent re-occurring injuries. This mindset involves regularly reviewing and updating safety practices. This includes incorporating feedback from safety audits, staying informed about industry best practices, and adopting new technologies or methodologies that enhance safety.

Preparing for emergencies and having a robust response plan in place are vital components of industrial safety. Effective emergency preparedness can mitigate the impact of incidents and ensure a swift recovery. Companies need to be ready, including when situations arise including, but not limited to:

- Explosion
- Fire
- Chemical spills, exposure
- Fall risks
- Equipment malfunction
- Unprotected openings
- Working on worker-lifts
- Spray pumps, hoses & clothing interfaces
- Fire-resistive material composition
- Much more…



Material and equipment storage is critical to safety during construction processes in existing buildings. Joe Riggi Photo

Emergency response plans should outline procedures for the various scenarios above, and much more. What's the response to fires, chemical spills, or natural disasters for the workers to implement? These plans should include clear instructions for evacuation, communication, and coordination with emergency services. Integrating the passive fire protection worker into the facility safety plan is critical to protecting workers against safety issues in the environment.

Regular training and drills are essential for ensuring that employees are familiar with emergency response procedures – again for both the facility and the passive fire protection materials and equipment used on the project site. Safety, evacuation and other drills should be conducted periodically and involve all employees to ensure preparedness. This is true for the passive fire protection workforce, and also the building manufacturing staff. Both should be aware of each other's risks for safety success to occur.

Ensuring that emergency equipment, such as fire extinguishers, first aid kits, and spill containment materials, is readily available and well-maintained is crucial. Regular inspections and maintenance of this equipment are necessary to ensure functionality during an emergency. It is important to note that equipment needed for facility operations might be different than the passive fire protection contractors' safety equipment.

CONCLUSION

Just stating the word "safety" brings varied meanings to anyone. "Safety Is…" could mean a class on reviving a stroke, heart attack victim or treating an injury. Safety in the construction and industrial operation industry does mean procedures to protect workers from injury. Staying ahead of safety in industrial occupancies involves a comprehensive approach that integrates regulatory compliance, technological advancements, employee training, and continuous improvement. By understanding and addressing the unique risks of industrial environments, organizations can create a safer workplace that protects employees and enhances operational efficiency.

As industrial, passive fire protection construction and maintenance operations continue to evolve, staying proactive and adaptive in safety practices will be key to managing new challenges and ensuring a safe working environment. Investing in safety not only safeguards employees but also contributes to the overall success and sustainability of industrial operations.

Joe Riggi, ASP, CRIS has been working in construction safety in the Boston area since 2014 working on projects ranging from heavy civil construction to ground up building construction in the commercial, healthcare and education fields. Joe obtained his CHST in 2019 and ASP in 2024. Since November 2021, he has been with G. Greene Construction as the Assistant Director of Safety. He can be reached at jriggi@ggreene.com

WRITTEN BY **WAYNE BARROW**

BARRIERS ARE THE HEART OF COMPARTMENTATION

Note: The common term of gypsum wallboard (GWB) is being replaced in this article by gypsum panels (GP) in keeping with the change to that term in the International Code Council's (ICC) International Family of Codes. "Gypsum panels" is already in use in some sections of the 2021 code and the Gypsum Association was a proponent of changing the terms in the 2024 and upcoming 2027 codes. This article is based on the 2021 International Building Code (IBC), part of the ICC family of Codes.

When you look at any UL Certified firestop system, the first component is the barrier. The barrier carries the fire-resistance-rating (F rating) and the firestop systems only restore the rating of the barrier where a breach of the barrier is created by the need for a penetrating item (ex: pipes, ducts, cables, etc.) or a joint between two adjacent barriers.

To me, the first step in firestop inspection is to verify that the barrier meets the requirements of the UL Certified wall design assigned to it. Then the rest of the firestop system must be verified to conform to the requirements set by the firestop manufacturer in the UL system listing. Especially in gypsum panel construction, we often see common deficiencies in the design and construction of vertical and horizontal barriers in Type V-A construction of multi-residential (IBC occupancy type R-2) projects. It is not unusual for the barrier deficiencies to be serious enough to diminish the effectiveness of the firestopping.

The most common construction type for multi-residential construction is Type V-A construction employing UL L500 series horizontal floor-ceiling assemblies. These floor-ceiling assemblies have three parts,

- A wood floor membrane
- An open-web truss structural framing. Engineered I-joists are sometimes used but less frequently.
- A fire-resistive gypsum panel (GP) ceiling membrane (Fig 1).

All three elements are tested as a unit and create a floor-ceiling assembly with a 1-hour F rating. It is incorrect to think of the subfloor as providing a 30-minute rating and the ceiling membrane as providing a 30-minute rating.



Figure 1: The entire floor-ceiling horizontal assembly carries the F rating. Continuity of the GP ceiling membrane is critical for protecting the trusses. Wayne Barrow Image

Walls are most often framed with nominal 2" x 4" and 2" x 6" wood studs and this article assumes all walls are single-stud width and are framed with wood studs only. The use of steel studs for interior partitions and walls constructed of dual rows of studs creates issues that need separate discussion.

FIRE PARTITIONS VERSUS FIRE BARRIERS: THE RATINGS ARE THE SAME

No matter which project delivery method is employed, a best practice is early engagement on firestopping by the design team and the general contractor (GC). This can be crucial to a project's success and should start with a detailed plan review no later than when the construction documents are 50% complete. Out of more than one hundred Type V-A projects we have reviewed, less than ten percent have had a code-compliant design of the


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walls between units. Many designers seemingly misunderstand the difference between Fire Partitions versus Fire Barriers and there is a mistaken idea that a Fire Partition is less robust than a Fire Barrier. However, the only difference between Fire Partitions and Fire Barriers is where the top of the wall terminates. Other than that termination point, both must be built in conformance with UL Certified wall designs which must meet the ASTM E 119 Standard Test Methods for Fire Tests of Building Construction and Materials or UL standard 263, Fire Tests of Building Construction and Materials. For example, UL Certified design #U305 has a 1-hour fire rating whether it is built as a Fire Partition or a Fire Barrier.

There are three critical International Building Code (IBC) prescriptive requirements that are often overlooked in the design of horizontal floor-ceiling assemblies and vertical barriers in Type V-A construction. These sections dictate the use of Fire Partitions.

IBC* section 711 deals with Floor and Roof Assemblies. Section 711.2.2 requires that "Assemblies shall be **continuous without vertical openings**, except as permitted by this section and Section 712." (emphasis added). The prohibition on vertical openings applies to all types of openings, not just large ones like stairs and shafts. However, since penetrations are required for mechanical, electrical, plumbing (MEP), and other service items, the IBC details how to deal with those primarily in Section 714.5 Horizontal Assemblies.

For the intersection of walls and horizontal floor-ceiling assemblies Section 714.5.2 Exception 7 states that "The ceiling membrane of a maximum 2-hour fire-resistance-rated horizontal assembly is **permitted to be interrupted with the double wood top plate** of a wall assembly that is sheathed with Type X gypsum wallboard, provided that all penetrating items through the double top plates are protected in accordance with Section 714.5.1.1 or 714.5.1.2 and the **ceiling membrane is tight to the top plates**."

So, rated and non-rated vertical partitions can interrupt the ceiling membrane by meeting those conditions. Though not common, non-rated vertical partitions can be built with a single top plate, but the GP ceiling membrane must continue uninterrupted above the top plate.

Last, IBC section 718.2.2 deals with all concealed wall spaces and says, "Fireblocking shall be provided in concealed spaces of stud walls and partitions, including furred spaces, and parallel rows of studs or staggered studs, as follows:

1. **Vertically at the ceiling and floor levels.**
2. Horizontally at intervals not exceeding 10 feet (3048 mm).

Put all three sections together and the IBC is clear that rated Fire Partitions (Fig. 2) that terminate at the ceiling level should be used Type V-A construction should be used rather than Fire Barriers that terminate at the subfloor. (Fig. 3).

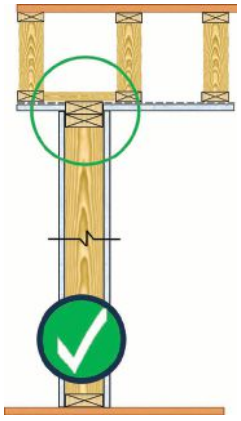


Figure 2: Fire Partition as required by the IBC with a double top plate & the ceiling membrane tight to the top plates

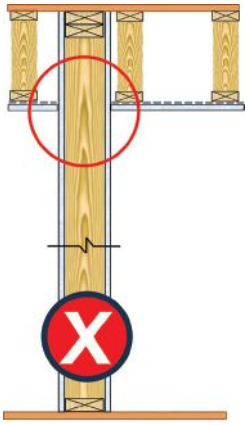


Figure 3: Incorrect "Fire Barrier" with no fireblocking

Wayne Barrow Images

This fireblocking at the ceiling level is critical for firestopping. Without fireblocking of at least a single top plate, there are no UL Certified firestop systems that allow protecting the openings around penetrations only at the subfloor, even if the penetrant is protected at the bottom of the framing and the subfloor level. (See Figs 4 & 5)

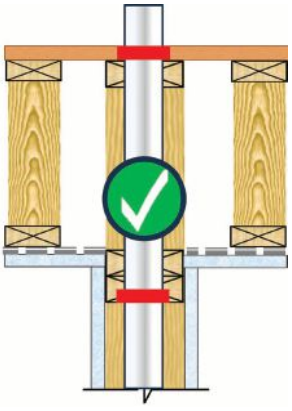


Figure 4: All through penetration systems for floor-ceiling assemblies require sealing at both levels.

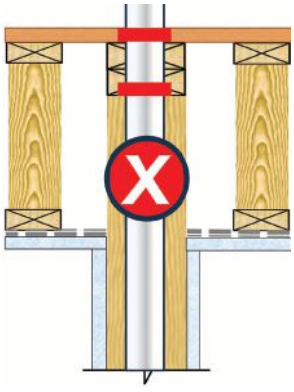


Figure 5: **NO** *through penetration firestop systems allow sealing just at the floor level.*
Wayne Barrow Images

A second issue involves the use of resilient channel (RC) on the demising walls that separate units. The use of RC will help increase the sound transmission class (STC) of a wall by creating a partial break between the framing and the GP. However, it is commonly mis-installed with the top row of RC installed at the plane of the GP ceiling membrane. That creates a ½" wide vertical opening for the entire length of the wall (Fig 6). Correct installation of RC places the top row of it just below the ceiling membrane so the ceiling membrane butts tightly against the top plate and covers the ½" gap between the GP wall membrane and the studs (Fig 7).

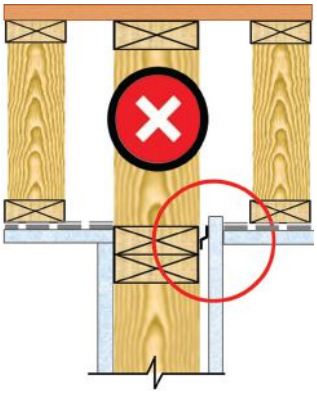


Figure 6: Incorrect installation of RC in the plane of the GP ceiling membrane and the resulting vertical opening.

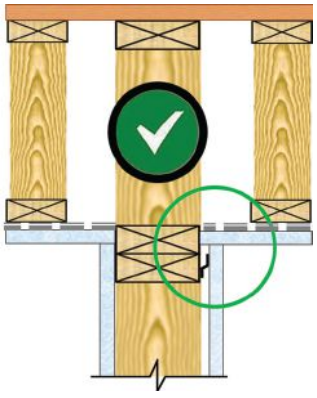


Figure 7: Correct installation of RC with the GP ceiling membrane extending to the top plates. Wayne Barrow Images

Two other common issues are when the GP ceiling membrane does not extend fully to the top plates at non-rated interior partitions and when the GP ceiling membrane does not extend across the top of chases that are often located in bathrooms for running DWV stacks and ducts. In Figure 8, the entire ceiling membrane in this area had to be removed and reinstalled. Commonly referred to as a "chase lid," missing GP in situations as shown in Figure 9 requires time-consuming rework. It is often best to install chase lids during initial framing and require each subcontractor to create their holes as needed.



Figure 8: Incorrect installation of the GP ceiling membrane above a shower stall. The gaps to the top plates were 2" to 3" wide.



Figure 9: Major time-consuming rework is required in this MEP chase because the GPceiling membrane is missing. Wayne Barrow Images

For fire resistance-rated GP assemblies three critical design and installation factors frequently create issues for achieving effective compartmentation.

First is the inclusion in the partition type details of appropriate UL Certified wall designs. To become certified every GP wall must meet the rigorous requirements of ASTM standard E 119 or UL standard 263. Both standards require stringent conformance to flame migration, temperature rise, and structural integrity limits. To meet those requirements the assemblies must be carefully constructed with all the requirements for stud type, stud sizing, stud spacing, screw type, screw spacing, and panel brand and type, panel alignment, and finish level fully met.

During plan review, one issue we often see is wall design details in the architectural plans that are generic with no UL Certified wall design assigned to them.

A second issue is plans with UL Certified wall designs listed that are inappropriate for the type of wall needed. On a recent multi-residential high-rise project, the UL Certified design listed for the corridor walls was a design for an exterior wall requiring steel panels on one side yet no one had noticed it even though drywall installation was starting.

A third issue is when a change is made to the UL Certified wall design, but no additional guidance is provided for installation. An example of this is from another multi-residential project when the design called for a UL Certified design that did not allow for adding resilient channel on either side. The architect required resilient channel which significantly decreased the screw spacing required by the original design. When consulted, what the GP manufacturer required was substantially different than what the drywall contractor was installing which could have allowed premature failure of the Fire Partition during a fire.

Even when these design issues are avoided, drywall contractors often do not follow the UL Certified wall designs. It is very common to see screw spacing on panel edges and in the field where the on-center screw spacing is half to one-third less than what is required by the UL Certified wall design.

Another common installation issue is gaps between the edges of the gypsum panels and patch edges. IBC section 2508.3 requires that "Edges and ends of gypsum board and gypsum panel products shall be

in moderate contact…" in fire-resistance-rated walls and floor-ceiling assemblies. Moderate contact is defined as having the panels touching fully along the edge without being forced together. The only gaps allowed are those from the manufacturing tolerances of gypsum panels which are limited to 3/32" across the width of the panel. Those variances will create very slight slivers of non-contact, not the large gaps seen in Figures 10 and 11. Gaps such as these cannot just be covered over by drywall compound and tape. The panels must be re-installed to meet the moderate contact requirement.

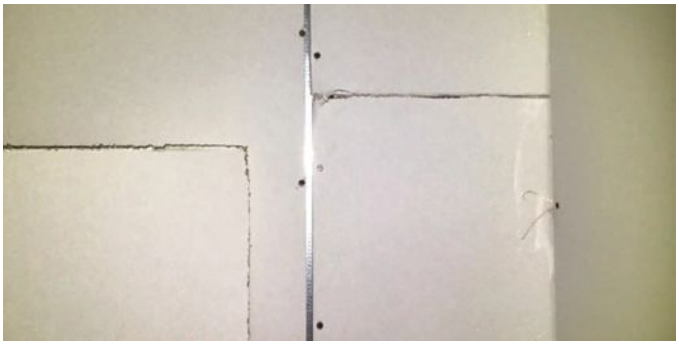


Figure 10: Even though the vertical gap is backed by a stud, these panels must be reinstalled, not just filled with tape and mud. Wayne Barrow Photo



Figure 11: Large horizontal gaps are especially serious because they are not backed by any framing. Wayne Barrow Photo

The heart of effective compartmentation is the vertical and horizontal fire resistance-rated barriers required by the IBC. To achieve effective compartmentation, those barriers must first be designed correctly and constructed correctly. Without correct design and construction, the barriers themselves will not provide the needed protection even if the firestopping of breaches at joints and penetrations is correct.

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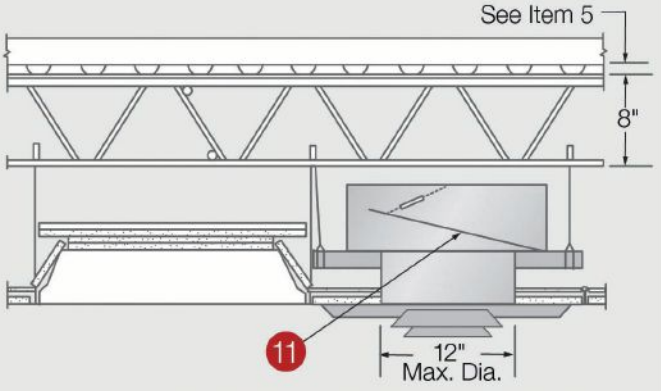
WRITTEN BY **MICHAEL J. BULZOMI**

ALL CEILING RADIATION DAMPERS ARE NOT EQUAL

The International Building Code (IBC), Chapter 7: *Fire and Smoke Protection Features*, Section 717.3.1 Damper Testing, lists the applicable test standards for each life safety damper type. Ceiling radiation dampers shall comply with the requirements of UL 555C *Standard for Safety, Ceiling Dampers* or shall be tested as part of a fire-resistance-rated floor or roof/ceiling assembly in accordance with ASTM E119, *Standard Test Methods for Fire Tests of Building Construction and Materials* or UL 263, *Standard for Fire Tests of Building Construction and Materials*.

UL 555C governs the requirements and methods of tests that apply to ceiling dampers and ceiling air diffusers intended for installation in hourly fire-resistance-rated floor and roof/ceiling assemblies.

UL 555C listed ceiling radiation dampers are approved for installation in assemblies that specify the use of a hinged door type dampers. We most commonly see these dampers in applications such as air inlet/outlet membrane penetrations located in fire-rated, lay-in acoustical tile ceilings in a non-combustible (steel) truss system fire-resistance-rated for 3 hours or less. See Figure 1.



11. Damper — No. 13 MSG galv steel, 14 by 14 in. Protected on both surfaces with 1/16 in. ceramic fiber paper and held open with a Fusible Link. (Bearing the UL Listing Mark). Damper to overlap duct outlet 1 in. min.

Figure 1. Example Diagram and damper description for an assembly permitting the use of UL 55C approved dampers.

A UL 555C listed ceiling radiation damper may also be tested in accordance with ASTM E119 or UL 263 and listed for use in a specific floor or roof/ceiling fire-resistance-rated assembly. ASTM E119 or UL 263 listed dampers are most commonly used in applications such as low-rise multi-family apartments, hotel and motels, assisted living facilities, dormitories, and residence halls, in gypsum board ceilings in combustible (wood) truss or joist system fire-resistance-rated for 1 hour. See Figure 2.



Figure 2. Ceiling radiation damper in a combustible wood-truss floor /ceiling assembly. Greenheck Image

Ceiling radiation dampers that have been evaluated and listed for use in a specific combustible wood truss or joist system are only intended for use in that design. In order to be qualified for use in other floor or roof/ceiling assemblies, additional testing and/or engineering evaluations may be required by the listing agency in order to be included in additional assembly designs. The damper manufacturer's installation instructions should list all the applicable floor or roof/ceiling designs that a specific model damper has been approved for – as the instructions incorporate the listings.

APPLICATION INSIGHT

While it may be true that the damper that was tested in accordance with ASTM E119 or UL 263 in a specific floor or roof/ceiling assembly and is also a UL 555C listed damper, you cannot substitute a UL 555C damper for the listed damper in a specific ASTM E119 or UL 263 floor or roof/ceiling assembly unless that damper has been evaluated for use in that specific assembly.

The damper listed in an ASTM E119 or UL 263 assembly is just a part of the complete rated assembly – which includes the damper, duct, trusses, gypsum panel, insulation, etc. – all the components of construction listed in the ASTM E119 or UL 263 floor or roof/ceiling assembly. The fire-resistance ratings apply to entire tested assemblies, as opposed to assemblies' individual components and materials. As such, ceiling dampers themselves are not fire-resistance-rated; instead, the ratings ceiling dampers carry indicate the dampers' suitability for use in specific rated floor/ceiling and roof/ceiling assemblies as a system.

LISTINGS & APPROVAL AGENCIES

We often see architectural specifications that reference a specific UL floor or roof/assembly and it is not uncommon for specifications to reference the Gypsum Association's Fire Resistance Design Manual or Warnock Hersey (ETL) listed dampers.

The Gypsum Association is not a testing agency; its listings are based on test data from various accredited third-party laboratories, all of which conduct fire tests of floor/ceiling and roof/ceiling assemblies in accordance with ASTM E119 or UL 263. The ceiling-damper models approved for use in a specific Gypsum Association design can be determined by consulting the test reports and/or design numbers of the testing agency listed in the design manual.

Warnock Hersey listed dampers are tested by Intertek (ETL) in accordance with applicable ASTM and UL standards as required by the IBC. As such, Intertek maintains a product directory that details the damper manufacturer, applicable test standards, and applicable UL floor or roof/assembly approvals. It is important to note IBC does not have a specific listing agency requirement for life safety dampers. Chapter 7, Section 717.3.1 states that dampers shall be "listed and labeled in accordance with" the applicable referenced ASTM and UL standard. It is at the discretion of the Authority Having Jurisdiction (AHJ) to accept and approve damper installations as well as the approval agencies that test and list the product.

DESIGN & INSTALLATION CONSIDERATIONS

The installation requirements for ceiling radiation dampers are outlined in the manufacturer's approved installation instructions for that specific damper model and are not typically found in the listing agency's floor or roof/ceiling assembly listing. However, the assembly listing will detail other critical damper related design and installation requirements. Failure to adhere to the assembly listing details and/or manufacturer's installation instructions could result in the improper operation of the damper and void the fire-resistance-rating of the assembly.



Ceiling radiation damper in a factory supplied plenum box. Greenheck Image

When specifying and installing ceiling radiation dampers that have been evaluated for use in 1-hour fire-resistance-rated combustible wood truss (or joist) floor or roof/ceiling assemblies, consider the following:

- In most applications, ceiling radiation dampers require a steel or fiberglass plenum (also referred to in the industry as a "boot", or a "box"). These plenums have specific construction requirements that can either be factory provided, or if approved by the listing agency, field fabricated. Damper manufacturers will outline field provided plenum construction requirements. Some manufacturers may also require the field installation of insulation over their factory provided plenums, and this is needed to maintain the listing of the product.

- The majority of floor or roof/ceiling assemblies are static rated assemblies. And for most static rated assemblies, the type of approved air duct is any UL Class 0 or Class 1 flexible air duct installed in accordance with the instructions provided by the damper manufacturer. Reference specific ceiling assembly details for duct requirements.

- If evaluated and approved, specific manufacturers' ceiling fan/damper assemblies or dampers designed to integrate into a ceiling fan, will be specified in the listing agency's assembly listing. Refer to the manufacturer's installation instructions for additional installation requirements.

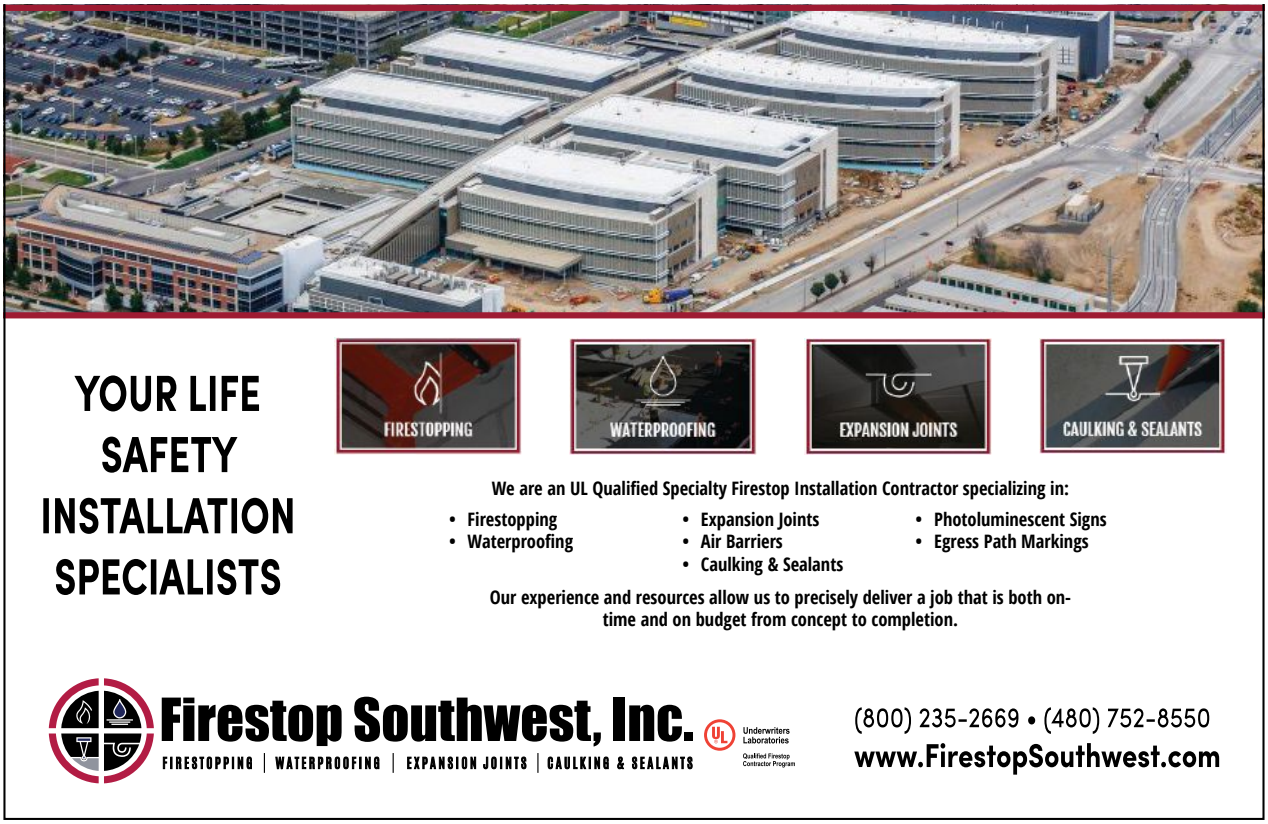
- For ceiling design and coordination between trades, it is important to note that aggregate damper openings are specified in the assembly listing, often expressed as "shall not exceed xxx sq. in. per 100 sq. ft. of ceiling area." This limitation will have an influence on membrane penetration placement within the ceiling.

- The architect and engineer should be aware of any air outlet/inlet sizing requirements as the available size range (max damper opening) can vary by damper manufacturer, limited to what was tested and approved. If there is not an approved damper large enough for a penetration, the penetration breach size should be redesigned. A UL 555 fire-resistance-rated horizontal fire damper has not been tested and approved for installation in a wood ceiling. A steel grille shall be installed in accordance with installation instructions.

- If possible, balancing dampers should be located at branch ducts. If balancing is required at the face of the air outlet, an optional volume control mechanism built into the ceiling radiation damper is recommended. There are limitations on how high damper blades can close above the ceiling, ranging from <1" to 3+" above the gypsum panel depending on manufacturer and model. The total depth of a grille with an opposed blade damper may be too deep to conform to the manufacturers' approved installation instructions once installed inside the required plenum. Consult damper manufacturer for damper plenum restrictions when using grilles with opposed blade dampers.

Ceiling radiation dampers are a critical fire and life safety device. They are required to be installed per manufacturer's installation instructions and product listings in accordance with the code adopted by the jurisdiction.

Michael Bulzomi is Product Manager, Commercial Dampers, at Greenheck, Inc. He can be reached at Michael.Bulzomi@Greenheck.com.



WRITTEN BY **LORI GREENE**

SPECIAL LOCKING ARRANGEMENTS VS NORMAL LOCKING ARRANGEMENTS -WHAT'S THE DIFFERENCE?

EDITORS NOTE: Where the article states 'I-Codes', it refers to the International Building Code (IBC) AND the International Fire Code (IFC). If not, the individual IBC or IFC is cited.

Electrified door hardware has become more common in every type of building. Whether it's a battery-operated lock on an apartment entry door, a controlled egress lock in a memory care facility, a delayed egress lock on a retail exit, or a fail-safe lock used for stairwell reentry, doors with access control hardware are becoming part of our daily lives.

The model codes address various types of special locking arrangements; these are typically electrified hardware applications that could affect egress in some way. But the most common type of electronic access control is not a special locking arrangement. On these doors, instead of allowing access with a metal key, access is controlled with a reader and electronic credential, and egress is not affected. This application could be considered a "normal locking arrangement."

If the term "normal locking arrangement" is unfamiliar, that's because it is not a term that is defined in the model codes. Past editions of the codes have not included a specific code section addressing doors with electrified hardware that allows free egress. These doors are intended to meet the same requirements that apply to doors with standard mechanical hardware. For example, if a door has mechanical panic hardware, it must allow free egress by pushing the touchpad of the panic hardware. Electrified panic hardware is subject to those same requirements and would allow free egress regardless of the status of the access control system.

When projects include electrified hardware that allows free egress while controlling access electronically, many code officials have referred to the code sections for special locking arrangements, looking for the applicable requirements. However, these sections do not apply to normal locking arrangements.

SPECIAL LOCKING ARRANGEMENTS

The sections in the 2024 model codes that are typically considered special locking arrangements are summarized below. The references in each section are from NFPA 101 – Life Safety Code and the International Building Code (IBC) and International Fire Code (IFC); together these are known as the I-Codes.



A delayed egress lock, which typically delays egress for 15 seconds while sounding an audible alarm, is one type of special locking arrangement addressed in the model codes. Allegion Photo

- **Stairwell reentry** applies to the hardware on the stair side of interior stairwell doors, which unlocks during a fire to allow building occupants to leave the stairwell if necessary and wait for assistance or find another exit. The electrified hardware used on these doors must be fail safe, so that it will unlock upon power failure, activation of the fire alarm or sprinkler system, and via a signal from the fire command center or other approved location. These requirements do not apply to the stair discharge door. 2024 I-Codes – 1010.2.6, 2024 NFPA 101 – 7.2.1.5.7

- **Door hardware release** applies to electrified locks (typically electromagnetic locks) that are released for egress via a switch in the door-mounted hardware, such as panic hardware, a lever handle, or a sensor bar with a request-to-exit (RX) switch. Upon normal operation of the door-mounted hardware, the mag-lock releases to allow egress. These locks are always fail safe – removing power will de-energize the electromagnet and unlock the door, as required by the model codes. 2024 I-Codes – 1010.2.10, 2024 NFPA 101 – 7.2.1.6.3

- **Sensor release** addresses electrified locks (typically electromagnetic locks) that are released for egress when a presence sensor detects a building occupant approaching the door. In addition to the sensor, the locks must be released by an auxiliary switch beside the door, upon activation of the fire alarm or sprinkler system, and upon loss of power. 2024 I-Codes – 1010.2.11, 2024 NFPA 101 – 7.2.1.6.2

- **Delayed egress** locks prevent a building occupant from exiting for 15 seconds under normal operation – or 30 seconds where approved by the authority having jurisdiction (AHJ). Delayed egress locks must allow immediate egress upon activation of the fire alarm or sprinkler system, upon loss of power, and upon a signal from the fire command center and other approved locations. Signage and an audible alarm are also required. 2024 I-Codes – 1010.2.12, 2024 NFPA 101 – 7.2.1.6.1

- **Controlled egress** locks in health care facilities are allowed to prevent egress until emergency evacuation is needed, but clinical staff must be able to release the locks at any time. The model codes include additional requirements for releasing these locks in an emergency, although there are exceptions for certain occupancy types. 2024 I-Codes – 1010.2.13, 2024 NFPA 101 – 18.2.2.2.5.3/19.2.2.2.5.3

- **Elevator lobby** doors may be electrically locked with fail safe locks to prevent unauthorized access to a tenant space, depending on the adopted code. Until the 2024 edition, the I-Codes did not address the special locking arrangements for these doors, but a new section has been added. One of the requirements for this application is a two-way communication system in the elevator lobby. 2024 I-Codes – 1010.2.14, 2024 NFPA 101 – 7.2.1.6.4

The model codes include many other requirements for special locking arrangements, to help maintain a balance of life safety along with security. Because the egress methods for these systems differ from typical hardware (i.e. panic hardware or latchsets/locksets), there are prescriptive mandates stating how this special hardware must function.

Note: Many of the sections above include a requirement for the electrified hardware to be listed to either UL 294 – Standard for Access Control System Units or UL 1034 – Standard for Burglary-Resistant Electric Locking Mechanisms (this alternative is new in the 2024 codes). However, these listings are not required by the model codes for the normal locking arrangements addressed below.

NORMAL LOCKING ARRANGEMENTS

As mentioned previously in this article, electrified hardware that controls access but does not affect egress is not considered a special locking arrangement. For these openings, the door hardware allows immediate egress regardless of the status of the access control system. These systems have been unofficially called normal locking arrangements, because for egress purposes they function like normal doors.



When a door has an electronic credential reader to control access, with hardware that allows free egress, this is not considered a special locking arrangement. Allegion Photo

Prior to the 2024 editions, the model codes have not included a specific code section for the type of electrified hardware that controls access but allows free egress. This hardware is required to comply with the same code requirements that apply to mechanical hardware:

- The door must unlatch with one releasing motion for egress (with some exceptions).
- Egress must require no key, tool, special knowledge, or effort.
- Operation of the door hardware must require no tight grasping, pinching, or twisting of the wrist.

- Operable force for the hardware must be within the limits of the applicable code or standard.
- Releasing hardware must be mounted between 34-48 inches above the floor (with some exceptions).

The lack of a model code section specific to this type of electrified hardware has caused confusion regarding which requirements should apply to these normal locking arrangements. However, a change was made to the 2024 edition of the I-Codes that will help to clarify the intent of the codes.

Beginning with the 2018 edition, the I-Codes have included a section addressing doors with electrified hardware that monitors or records when the door is used for egress. Beginning with the 2024 editions of the IBC and IFC, this section will also apply to doors with access control systems:

1010.2.9 Monitored or recorded egress, and access control systems. Where electrical systems that monitor or record egress activity are incorporated, or where the door has an access control system, the locking system on the egress side of the door shall comply with Section 1010.2.11, 1010.2.12, 1010.2.13, 1010.2.14 or 1010.2.15 or shall be readily openable from the egress side without the use of a key or special knowledge or effort.

This section states that when a door is equipped with one of these types of systems, the electrified hardware can **EITHER** comply with one of the sections addressing special locking arrangements **OR** the door must be readily openable from the egress side without the use of a key or special knowledge or effort. This section addresses doors with electrified hardware that controls access but does not affect egress, and clarifies that when a system allows free egress, it does not have to comply with the sections addressing special locking arrangements.

Annex A of NFPA 101-2024 has also been modified, to clarify the intent of the code regarding access control systems that allow free egress. The annex note applies to paragraph 7.2.1.5.3 which addresses the latch releasing devices (door hardware), to make sure that the hardware on the egress side of the door that has an obvious method of operation and that is readily operated under all lighting conditions.

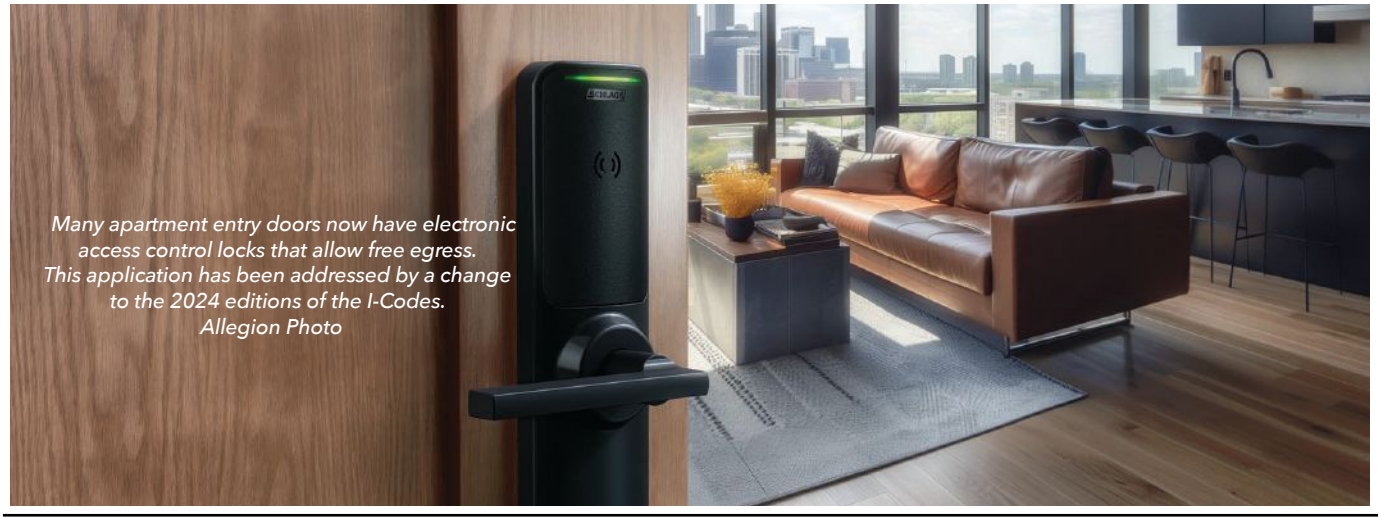
The new note in NFPA 101-2024 Annex A states in part:

A.7.2.1.5.3 - …Also common are electrical access control systems such as an electric lock or electric strike controlled by the input of a card reader or keypad. On the egress side of these doors, the lever-operated hardware, panic hardware, or fire-exit hardware must provide for immediate and unimpeded egress…

This clarifies that latch releasing devices might include electrified access control systems, with hardware that provides free egress. These changes to the 2024 model codes should help to ensure proper interpretation and application of the requirements.

For more information about special locking arrangements and electrified hardware, refer to iDigHardware.com/SLA. Reference the adopted codes for the detailed requirements applicable to a project's jurisdiction and connect with the AHJ if assistance is needed.

Lori Greene, DAHC/CDC, CCPR, FDAI is Manager, Codes & Resources at Allegion. Lori has been part of the door and hardware industry for more than 35 years. She currently acts as a resource for Allegion employees and customers in the area of building, fire, and accessibility codes as well as specifications. She shares information on these topics each weekday on her website, iDigHardware.com. She can be reached at Lori.Greene@allegion.com.

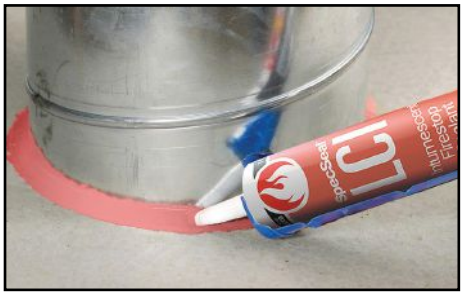


Many apartment entry doors now have electronic access control locks that allow free egress. This application has been addressed by a change to the 2024 editions of the I-Codes. Allegion Photo

SpecSeal® LCI Intumescent Sealant: The Workhorse of the Firestopping World

Selecting the right firestop sealant is essential. Intumescent, water-based latex sealants aren't just extremely versatile, they are also safe, non-toxic, and clean up easily with water. Intumescent sealants are the workhorse products of the firestopping world. And SpecSeal® LCI Intumescent Sealant is the tried-and-true firestop that contractors depend on to get the job done right the first time.

SpecSeal® LCI Intumescent Sealant is a premium-grade intumescent that stays flexible over time to accommodate normal building movements and pipe vibration. Available in conventional 10.1 oz (300 mL) caulk tubes, 20 oz (592 mL) foil-packs (i.e., sausages), and 5-gallon (19 L) pails, it can be easily caulked, bulk-loaded, or troweled into openings in fire-rated floors and walls.



Typical Concrete Floor Application for Single Pipe

When heated to 350F (177C), SpecSeal® LCI Intumescent Sealant expands volumetrically up to 10 times to form a dense, hard packed char. LCI is a great product for metal pipes, conduits, or tubes and is tested for a range of metallics, including steel, iron (including asphaltic and epoxy-coated), copper, and aluminum. Pipes up to 30" are tested and approved. LCI has many tested and listed systems available for up to 2" trade size (50 mm) when dealing with PVC, CPVC, and PEX pipes, providing multiple solutions. Anything above 2" will sometimes require accessory products such as wrap strips or collars. Certain critical plastics may also require other accessory products so check the pipe type printed on the side wall of the pipe.

LCI Sealant is a great solution for up to 2" (50 mm) thick glass fiber pipe insulation, 1" (25 mm) AB/PVC insulation (Armaflex, K-Flex, and Rubatex are common trade names), 3" (75 mm) cellular glass (Foam-Glas or equivalent), and 3" (75 mm) mineral wool pipe insulation. Insulation thickness above these ranges may require other accessory items such as intumescent wrap strips. Beware that there are other types of foam insulations that may be problematic. If in doubt, STI's Engineering Services are only a phone call away at 800.992.1180.

SpecSeal® LCI Intumescent Sealant is tested on non-energized busways: remember, water-based sealants and energized busways don't mix. LCI is also tested for round, rectangular, and flat oval ducts through all common constructions. Rectangular ducts where a dimension exceeds 24" may require additional steel retaining angles (L-shaped) to reinforce the ductwork and prevent it from deflecting under fire conditions. If a fire damper is present, call STI Engineering Services at 800.992.1180 for specific instructions on how to proceed.

Complex openings with multiple or mixed penetrants can also be safely sealed with LCI Sealant. In wood or steel-framed walls, the opening may need to be framed in and packing material may be appropriate depending on the tested and listed system.



Typical Gypsum Panel Wall Application for Single Penetrant



Typical Head-of-Wall Detail

The common joints sealed with LCI Sealant are head-of-wall applications between gypsum board/steel framed walls or concrete/masonry walls and concrete slab floors or steel deck assemblies. LCI can be used for these applications, but if greater movement capabilities are needed, other STI sealants like SpecSeal® ES Elastomeric Firestop Sealant may be employed that offer greater elasticity and range of movement. Where the steel ceiling runner is in the plane of the joint, sometimes a bond breaker tape or paraffin wax can be applied to the steel ceiling runner to prevent three-sided adhesion and ensure better movement characteristics.

With such versatile application offerings, SpecSeal® LCI Intumescent Sealant is truly the Workhorse of the Firestopping World. Reach out to your local STI Rep to learn more about LCI and all of their innovative firestop solutions.

WRITTEN BY **FCIA STAFF**

THE 'I'-INSPECTION AND FCIA'S 'DIIM'™

Firestopping is a very technical, detailed installation, very different from other construction disciplines. FCIA built the Firestopping 'DIIM'™ to provide general contractors, firestop contractors, code officials, specifiers and inspection agencies with tools to result in quality installations.

'**D'-Design** - Specification agencies using CSI/CSC MasterFormat Numbering System, tend to categorize the discipline in 07 84 00 Firestopping. This section keeps all firestopping in one place, for concise communication of code and environment condition requirements to the contractors through construction documents. We know manufacturers spend millions testing products for suitability for use in specific applications – to build fire-resistance designs. And, our specifier friends have included the 'DIIM'™ into specifications worldwide.

'I'-Installation – In the FM 4991, Standard for the Approval of Firestop Contractors, and UL Solution's Qualified Firestop Contractor Program, specialist firestop contractors are audited by FM Approvals or UL Solutions personnel, in the office and at the field. The program includes a person who must pass the FM or UL Firestop Exam based on the FCIA Firestop Manual of Practice, who is named a Designated Responsible Individual. Knowledge with quality process, means firestopping processes are managed. Field reports state that the 'I'-Inspection can be easier and faster when a FM 4991 Approved or UL Qualified Firestop Contractor is hired to install as a single source company.

'I'-Inspection – The ASTM E2174, ASTM E2393 Firestop and Joint Inspection Standards are the result of FCIA drafting a standard with a hired consultant in 2000. The FCIA Standards Committee then worked with ASTM consensus to publish the standard inspection procedures for firestopping - ASTM E2174/ASTM E2393. The standards were added to the 2012 International Building Code, NFPA 101 Appendix during the 2009/2010 code development cycle. Firestop special inspection became required for:

- High Rise Buildings – Defined as highest occupiable floor or roof that is 75' above lowest fire department access.

- Category III and IV Buildings in Table 1603.5. Schools with greater than 250 occupants, hospitals, police and fire stations, and other structures critical to safety fall in this category of building.

NEW RESIDENTIAL IBC FIRESTOP SPECIAL INSPECTION

In the 2018/2019 International Building Code Development Cycle for the 2021 International Building Code, FCIA was the proponent to add a new category of buildings that need firestopping special inspection. Residential Occupancies with greater than 250 occupants – similar to the school occupancies was added. This important addition to the 2021 International Building Code helps increase building safety through firestop special inspection.

Additionally, FCIA's proposals adding that installation is are in accordance with the *listing* and *manufacturers' instructions* have been successful, providing the acceptance criteria for firestop special inspection agencies to judge installed firestopping.

714 Penetrations

714.2 Installation. A *listed penetration firestop* shall be installed in accordance with the manufacturer's installation instructions and the listing criteria.

715 – Joints and Voids

715.2.1 List system installation. *Listed fire-resistant joint systems*, perimeter fire containment systems and *continuity head-of-wall systems* shall be securely installed in accordance with the manufacturer's installation instructions and the listing criteria in

or on the joint or void for its entire length so as not to dislodge, loosen or otherwise impair its ability to accommodate expected *building* movements and to resist the passage of fire and hot gases.

Most construction industry associations are against added inspection. FCIA is different. We know that the inspection process – performed correctly by competent inspection agencies employing inspectors – can increase likelihood that firestopping is installed in accordance with the listing and manufacturers' instructions. This means a safer building. At an FCIA Conference, Dennis Hall, FCSI, FAIA, stated at "You get what you Inspect" – first, in 2006 – supporting safer buildings.

INSPECTION BY WHOM?

When the International Building Code requires firestop (or any other) inspection, the inspection agencies and employed inspectors need to be approved by the Authority Having Jurisdiction. To be approved, both the inspection agency and inspector shall be independent, objective, competent, and experienced in the same type and complexity of materials and systems inspected. We see this as evolving rapidly as jurisdictions implement procedures to approve firestop special inspection companies and people. Read on for how this happens….

'I'-Inspection Agencies can demonstrate competency, having inspectors, and equipment through IAS's Accreditation Criteria for Firestop Special Inspection Agencies - AC 291. Specifiers and building owners can quantify Inspection Agency qualifications using IAS AC 291. More on this in a future *Life Safety Digest* article.



Both Installation & Inspection – require manufacturers' installation instructions and listings for firestopping done right the first time. FCIA Photo.

Inspection Agency firms are to employ qualified inspectors to review contractor firestopping installations. Inspector qualifications include experience and competence.

- **Independent/Objective** – ASTM E2174/ ASTM E2393 require notarized conflict of interest statements that the inspection agency and inspector is objective, independent of the specialist firestop contractor, and not a competitor of the supplier or the contractor installing firestopping, nor the specialist firestop contractor. The IBC requires objectivity as well, as determined by the code official.
- **Competent** – The FM or UL Firestop Exam, IFC Exam, ICC Certificate of Learning Achievement, WABO Exam all provide proof of individual inspector competence. Ask for this as an AHJ, or purchaser of firestop inspection.
- **Experience** – Inspecting a small firestop installation project is different than inspecting a complex multifamily building, a semiconductor chip factory or hospital. The inspection agency and their inspectors need to have experience – in each type of building inspected - to understand how to analyze the firestopping compliance, or non-compliance.

'M'-Maintaining Protection for the building life cycle is critical to firestopping and passive fire protection in buildings, protecting against fire and smoke spread that can cause structural damage and loss of building use. Maintaining all Passive Fire Protection Features keeps fire and life safety from being compromised. More on this in a future *Life Safety Digest* article, as the 2027 IBC will have changes – that are going through the development process now!



Experienced Firestop Inspectors would reject. One wrap strip for a 12" PVC Pipe through a fire-resistance-rated wall. FCIA Photo

The FCIA's 'DIIM'™ Story for Firestopping did not happen by magic. It was purposefully established to build better fire-resistance in buildings by FCIA's Leadership. Check it out at www.FCIA.org/About-FCIA.

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INDUSTRY NEWS AND CODE CORNER

CANADA'S 1ST PasFiPro SYMPOSIUM



FCIA/NFCA Member Don Falconer

The Firestop Contractors International Association (FCIA) & National Fireproofing Contractors Association (NFCA) partnered to build "PasFiPro '24" in Calgary, Alberta, Canada. With over 130 in attendance, this joint Symposium brought two fire-resistance industries together to promote better fire and life safety in Canada through accurate codes, standards, installation, inspection and maintenance of fireproofing and firestopping. Great speakers from the National Research Council of Canada, Codes Canada, NFCA Members on ULC-S101, ULC-S115, tabletop exhibits and much more. Looking forward to PasFiPro '25!!



FCIA Platinum/ NFCA Silver Member, Hilti @ PasFiPro

WEBINARS

Many passive fire protection entities are providing webinars to educate about this important topic. Associations and manufacturers provide education – some generic, some with sponsored content. Regardless of type, it's great to see passive fire protection getting the word out about this important discipline. Visit www.FCIA.org – www.NFCA-online.org – https://doorsafety.com/ - www.UL.com and other websites to gain educational credits for accreditations, certifications. Some are AIA Accredited, ICC PPP's, or other certifying organization authorized.

FREE FIRE DOOR INFO

NFPA 80 Chair Keith Pardoe, DSC, DSI, FDAI, DAHC, CDC, CDT – www.DoorSafety.com leader, offers free education on fire-doors including flash cards for studying requirements for inspection and testing of swinging fire doors. Keith has been a *Life Safety Digest* contributor over many years.

APPROVALGUIDE.COM

FM Approvals' operates an approval guide for 26 categories of construction items, from firestopping to roofing, sprinklers and pallets. FM Approvals' investment in this resource has been extensive. Looking for listings for fire doors or explosion venting wall systems? Firestopping and FM 4991 Approved Firestop Contractors? Check out www.ApprovalGuide.com.

GYPSUM ASSOCIATION DOCUMENTS

The Gypsum Association released its 2024 versions of the GA-600, *Fire Resistance and Sound Control Design Manual* recently. With over 800 fire designs referenced, it is a good resource for gypsum panel based fire-resistance-rated assemblies. Free downloads available at the GA Website, www.gypsum.org.

INTERTEK LISTINGS

The Intertek directory has listings for fire-resistance-rated assemblies, firestopping, perimeter fire containment, doors, much more. There's a link from www.Intertek.com home page. Or, check out the searchable database.

UL PRODUCT iQ USES GROWING

We all know that the www.UL.com/PiQ website has listings for many industries, including fire-resistance-rated assemblies and their breach protection items – firestopping, fire doors, fire-rated glazing, fire dampers & structural fire-resistance. Did you know there are digital marketing opportunities available? Check out the Prospector Digital Marketing at UL Solutions.

PASSIVE FIRE PROTECTION @ ICC

The International Code Council's Expo & Committee Action Hearing #2 for development of the 2027 International Building and Fire Code Development Process took place in Long Beach, CA in October.

At the event, a joint program from the FCIA & NFCA on fire-resistance with over 130 attendees was presented by Bill McHugh & Consultant to NFCA & FCIA, Rich Walke. Great program, great expo booth, productive code development hearings.

During the Code Development Hearings NFCA's code team also had several proposals heard by the Fire Code Committee and Fire Safety Committee.

Fire-Code proposal F89-24 was approved. The proposal changed the name of *sprayed fire-resistant* materials to *sprayed* ***fire-resistive materials***. It also changed *intumescent mastic, intumescent coatings*, to ***intumescent fire-resistive materials.***

Get ready for new names in the International Fire Code (IFC) '27, that matches the International Building Code (IBC '24). Fire marshals agreed that IFRM's are fireproofing, not a paint or coating. Coating/paints seem to point to no listing required. IFRM means Fireproofing – install materials in accordance with the listing and manufacturer's instructions. This proposal moves to the consent agenda with a very high possibility of approval.





Consultant to FCIA/NFCA Rich Walke & Bill McHugh speak & testify on code proposals at ICC '24



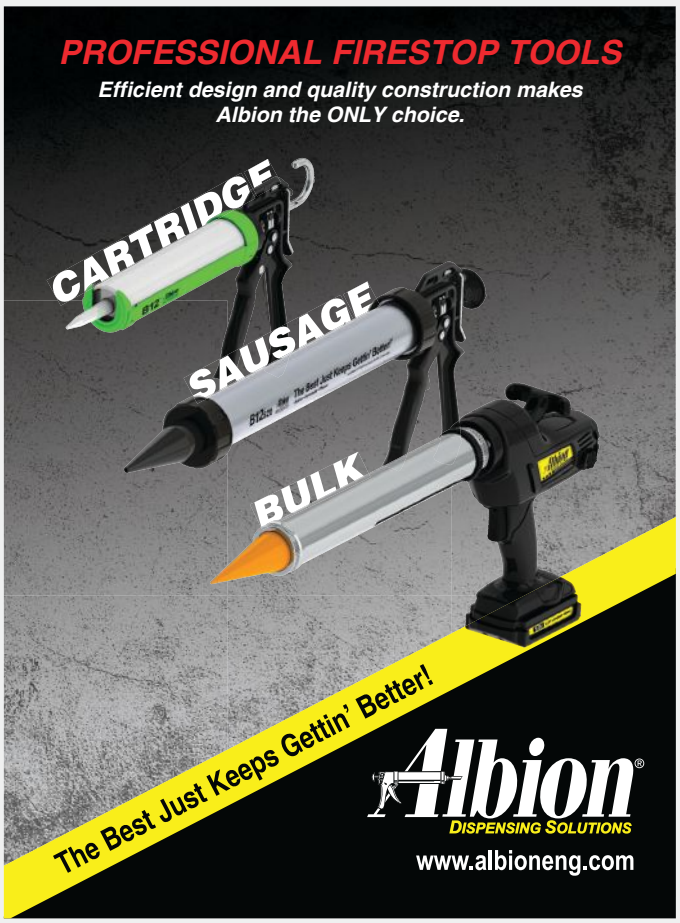
International Code Council Committee Action #2 Hearings. FCIA Photo

Another Fire-Code proposal that passed and was improved upon at the hearings provides guidance to fireproofing and general contractors, fire marshals, and code officials for fireproofing repairs. Repairs need to be performed "to the requirements of the listing and in accordance with the original manufacturer's instructions", was added and will be in the '27 International Fire Code.

For all these proposals, we will know final outcome after the International Code Council's Public Comment Hearing, April, 2026.

In Firestopping, FS47-24 kept firestopping in parking garages where cars are located adjacent to the penetration. **FS55-24** brought perimeter fire containment back to the roofline, to protect firefighters who are working on rooftops. In **FS42-24**, the proposal to add identification devices (labels) where penetrations and joints occur, did not make it through this hearing and will be disapproved. Objections were that the new ASTM Standard is not ready yet.

There's a lot more that goes on at the code hearings. Once the process finishes, we'll update with much more info on specific code proposals that passed through the complete process.



2025 FCIA INDUSTRY CALENDAR

JANUARY

January 14-16
Intersec Dubai
Dubai, UAE
www.intersec.ae.messefrankfurt.com

FEBRUARY

February 9-10
FCIA-NFCA PasFiPro Symposium
Doha, Qatar
www.fcia.org/Events/Conferences-Symposiums

February 12-13
FCIA-NFCA PasFiPro Symposium
Dubai, UAE
www.fcia.org/Events/Conferences-Symposiums

February 15-18, 24-27
Big 5 Saudi
Riyadh, KSA
www.big5constructsaudi.com

February 25-27
Associated Builders and Contractors Convention
Las Vegas, NV
www.abcconvention.abc.org

MARCH

March 9-12
ASHE PDC Summit
Atlanta, GA
www.ashemarketingsolutions.org

March 12-13
International Facility Managers Association (IFMA)
World Workplace Europe
The Hague, Netherlands
www.worldworkplaceeurope.ifma.org

March 30–April 3
AWCI Annual Convention & Expo
Charlotte, NC
www.awci.org/events/build-convention

March 31–April 4
NFCA Week of Learning
Broomfield, CO
www.nfca-online.org/events/index.asp

APRIL

April 7-9
ASTM E06 Meetings
Toronto, ON

April 8-10
Associate General Contractors of America Annual Convention
Columbus, OH
www.agc.com

April 10-12
Fire Safe Build India
Mumbai, India
www.fsbi.in

April 27–May 6
ICC Committee Action Hearings
Orlando, FL
www.iccsafe.org

April 29–May 1
IFMA Facility Fusion
Austin, TX
facilityfusion.ifma.org/

MAY

May 4-7
Riskworld
Chicago, IL
www.rims.org/annual-conferences/riskworld-2024/save-the-date

May 12-16
FCIA ECA '25 - Education and Committee Action Conference
San Antonio, TX
www.fcia.org/Events/Conferences-Symposiums

May 21-25
Construction Specifications Canada Conference
St. John's, Newfoundland
www.CSC-DCC.ca

JUNE

June 4-7
AIA Conference on Architecture
Boston, MA.
https://conferenceonarchitecture.com

June 16-18
NFPA Conference & Expo
Las Vegas, NV
www.NFPA.org

June 28–July 1
BOMA International Conference & Expo
Boston, MA
www.BOMA.org

JULY

July 27-30
ASHE Health Care Facilities Innovation Conference
Columbus, OH
https://www.ashe.org/education/health-care-facilities-innovation-conference

AUGUST

August 10-14
NASFM Annual Fire Prevention & Safety Symposium
Biloxi, MS
www.firemarshals.org

SEPTEMBER

September 5-8
Council on Tall Buildings and Urban Habitat International Conference
Toronto, ON
www.2025.ctbuhconference.com

September 14-16
Canadian Healthcare Engineering Society (CHES) Annual Conference
Calgary, AB
www.ches.org

September 14-17
RISKWORLD - Canada
Calgary, AB
www.rims.org

September 17-19
IFMA World Workplace
Minneapolis, MN
www.worldworkplace.ifma.org

OCTOBER

October 12-16
FCIA FIC '25 – Firestop Industry Conference & Trade Show
Orlando, FL
www.fcia.org/Events/Conferences-Symposiums

October 14-15
OFSEC
Muscat, Oman
www.ofsecevent.com

October 19-30
ICC Annual Conference, Expo, Committee Action Hearings
Cleveland, OH
https://www.iccsafe.org/about/news-and-events/annual-conferences-hearings/future-annual-meetings/

October 20-21
Expo Fire AMRACI
Guadalajara, Mexico
www.expofire.mx

NOVEMBER

November 10-14
AWCI Industry Leaders Conference
Louisville, KY
www.awci.org/events

November 25-27
FireEX
London, UK
www.anticipate-event.com/london/en/more/about/firex.html

FCIA-NFCA PasFiPro Symposium
Canada
Dates and Location to be announced.
www.fcia.org

Any event listed is subject to cancellation or date change.

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